

On Day 1, we started Lab 4 and worked on using the FPGA to speed up the GEMM matrix multiplication part of the digit recognition system. We got through the setup and made it up to Part 4, where we began changing the host code for the FPGA GEMM backend.

We edited the `gemm_kernel.c` file and tested it with the Jupyter notebook to compare our output with the CPU version. Some struggles we had were fixing row and column indexing, keeping track of M , k and N , and making sure the kernel name matched. By the end, we were working on creating buffers, sending data to the FPGA, setting kernel arguments, and reading the result back.